

GrantLens

System map — an eligibility-assessment assistant for grant applications. AI provides the evidence; people make the decision.

Version 2.8 · September 2026
AI core: Qwen3-8B, running locally
Rule sets: NSF CAREER (US) + 4 Australian programs + 1 demo

FIGURE 1

Architecture and data boundary

All processing — including the language model — stays on the organisation's own server. Applications are never sent to any external service.

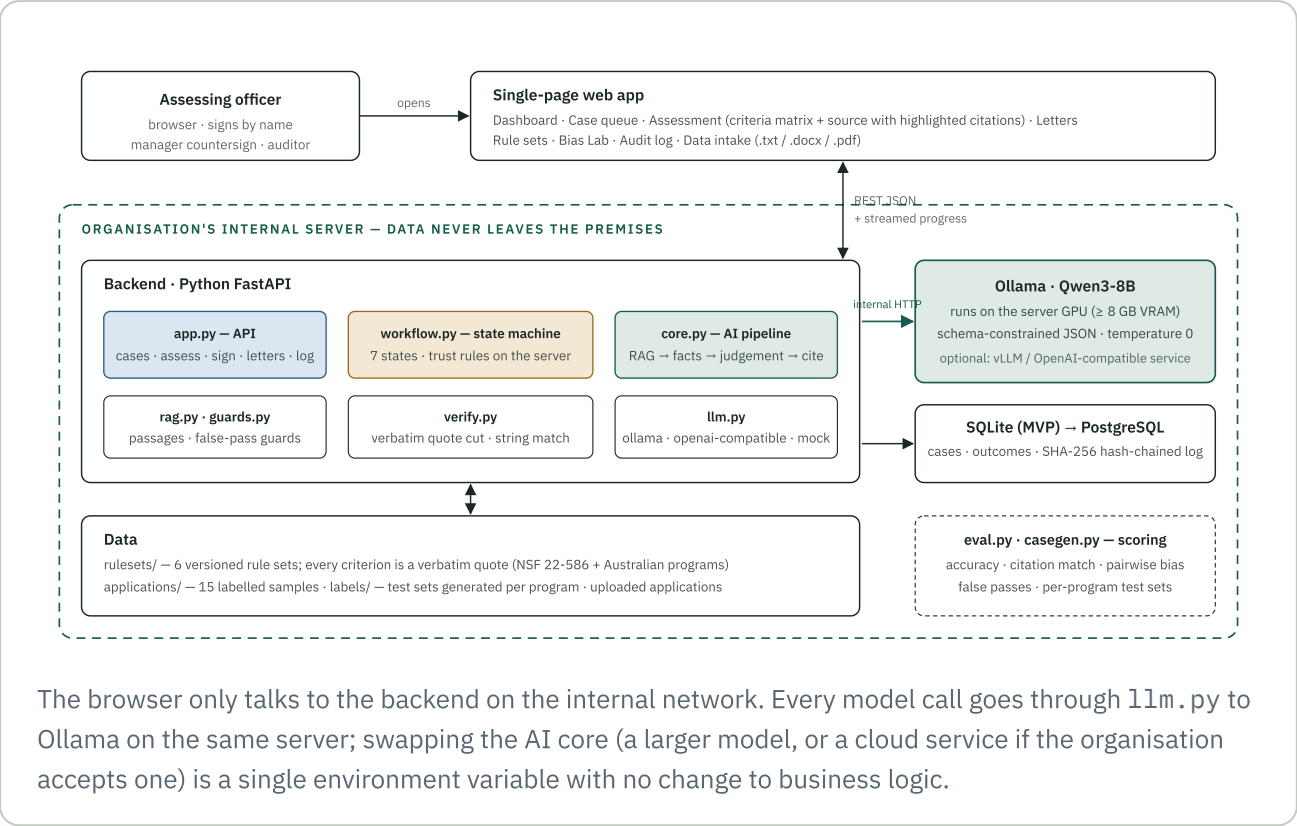
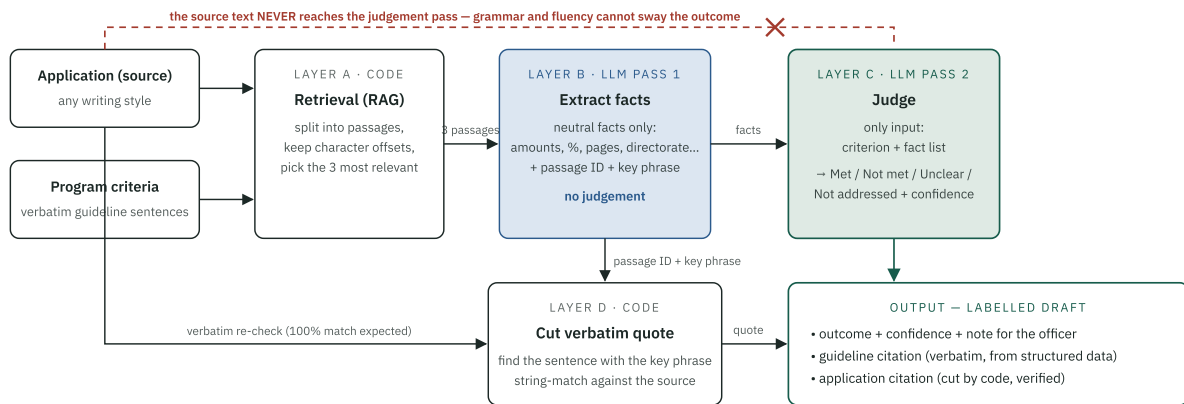


FIGURE 2

The AI pipeline for one criterion — why writing style cannot sway the outcome

Each criterion passes through four layers. The decisive point: the judgement pass only receives a normalised fact list and never sees the original text, and every quote is cut by code

rather than written by the AI.

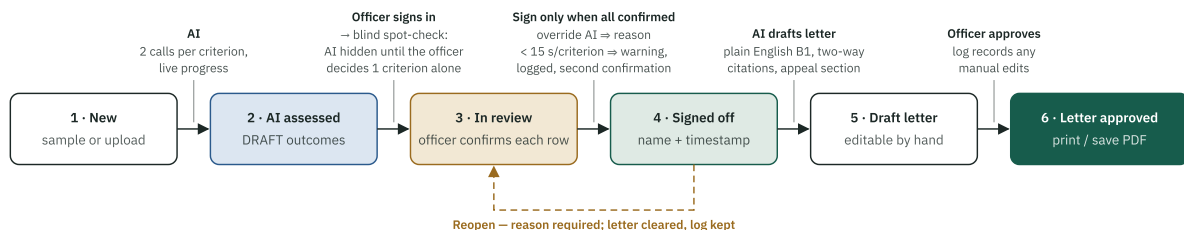


Two applications with the same facts but different writing styles produce the same fact list in layer B, and therefore the same outcome in layer C — this is what Bias Lab verifies on two pairs of sample applications. Layer D guarantees verbatim quotes by construction: the 8B model is never allowed to write a quote itself.

FIGURE 3

Case workflow — the controls sit on the arrows

Each application is a case moving through states whose transitions the server controls. Since v2.2 an **Awaiting supplement** state (multiple rounds, SLA deadline) sits between Review and Sign-off, and a **manager countersignature** is required before any rejection letter is issued.



Hash-chained audit log — written on every arrow

What the AI proposed · who changed what, and why · spot-check result · who signed, when · each event stores the previous event's SHA-256 — edit one line and the chain breaks

There is no "approve all" button: every criterion is a separate confirmation. An AI outcome of "Unclear / Not addressed" forces a human decision with a reason, and that reason flows straight into the outcome letter's "what to supplement" section.

MEASUREMENT

Quality, measured program by program

Every program has to earn its own figures. A result is published only once a program has completed the whole chain below — one program's figures are never used to describe another.

1 AI generates a test set: 1 clean application + applications with exactly 1 planted violation

2 Code fixes the violation, blocks answer leaks, rejects text that is not an application

3 An independent checker model (different family from the system under test) flags doubtful cases

4 A reviewer reads every application, confirms or discards

5 Labels approved

6 Measured with the real AI core: violations caught · false passes · false alarms

PROGRAM / RULE SET	CODE-GUARDED CRITERIA	MEASUREMENT STATUS	PUBLISHED RESULT
NSF 22-586 CAREER (US)	8 / 12	measured, full chain	caught 1/1 · false passes 0/1 · false alarms 0/1
Wine Tourism R8	5 / 9	measured, full chain	caught 2/2 · false passes 0/2 · false alarms 1/2 (earlier 3/3 figure withdrawn)
Cyber Security Skills R2	4 / 8	measured, full chain	caught 2/2 · false passes 0/2 · false alarms 0/2
Boosting Female Founders R1	3 / 6	measured, full chain	caught 1/2 · false passes 0/2 · referred to officer 1 · false alarms 0/2
On-farm Water	4 / 8	measured, full chain	caught 1/1 · false passes 0/1 · false alarms 0/1
Demonstration fund	1 / 3	measured, full chain	caught 1/1 · false passes 0/1 · false alarms 0/1

Why the earlier Wine figure was withdrawn: when we extended measurement to the other programs, we found that Wine's generated test documents leaked the answer (a test application stating "violating W08"), so the 3/3 figure is no longer valid. Test-set generation has been rebuilt with the checks in steps 2–4. **The re-run shows why step 4 is mandatory:** 18 AI-generated violation cases passed every automated check, yet the reviewer still discarded 9 of them (5 wrong labels, 4 ambiguous answers). **Sample size:** after review, 1–2 violation cases remain per program — results are preliminary per-program evidence, not statistical rates.

What the measurement found

All 6 programs (system under test: qwen3:8b): **8/9** planted violations caught · **0/9 false passes** · 1 case referred to an officer · clean applications decided correctly 8/9 · **1/9 false alarm**.

The guard layer stopped a real AI false pass. In Female Founders F05 the application plainly states "registered income tax exempt entity" — an excluded category — yet the model marked it Met. The code guard caught it, downgraded it to Unclear with a warning, and required an officer decision. This is the only case referred to an officer; no false pass got through.

One false alarm, with a known cause. In Wine W06 the clean application states "operates a dedicated physical cellar door located on our property", but the model returned Not met because "there is no information about ownership or lease" — missing information should have been Unclear. Added to the improvement backlog: a code-level check that a Not met outcome must be backed by a violating fact. That change will be re-measured on all six programs before it is published.

SEPTEMBER 2026 UPDATE — FROM THE CLIENT REVIEW

Added mechanisms (implemented and tested)

Never leave the AI alone where code can't check

For every criterion code cannot verify (including purely qualitative ones), a Met outcome **requires a genuine verbatim passage from the application**. Code checks it, so typing enough words won't pass, and pasting the same text across criteria is blocked; the attestation is logged under the signer's name.

Open work, published by priority

Wherever code cannot check yet, the system **lists it as a work item** (API /api/guard-backlog), ranked by risk: quantitative criteria without a guard = high, partial coverage = medium, purely qualitative = low. **All high-priority items are closed across all 6 rule sets**; conditional exclusions (e.g. "100 or more employees and not compliant with the Workplace Gender Equality Act") are now listed as gaps instead of being silently skipped.

Limitations published, not hidden

Every rule set has a **3-level coverage report**: code-guarded / **needs a hand-written guard** (derived logic, alternative branches, conditional thresholds or exclusions) / model + officer only. A false-pass figure is only valid for **the program measured, on a test set with approved labels**, and the result must match that exact label set — otherwise the system shows "re-measurement required".

Multiple grant opportunities, versioned

Each round or program has its own versioned rule set; an application **locks a snapshot of its rule set at first assessment**, so changing rules mid-round never affects cases in progress. Four real Australian programs are loaded, quoted verbatim from their guidelines: Wine Tourism R8, Cyber Skills R2, Boosting Female Founders R1, On-farm Water — covering all four program types.

Code guards against false passes

Criteria measurable in numbers (budget, %, page counts, co-PIs, cellar door, joint applications, 51% women-owned...) are re-checked by code after the AI: if the AI says Met but the figures breach the rule, the outcome is overridden to Not met and logged. Measured with real Qwen on 2 NSF sample applications containing every trap (6 unmet criteria): **false passes fell from 1/6 to 0/6, accuracy 24/24.**

Countersignature · supplement rounds · SLA

Applications with an unmet or unclear criterion only receive a letter after **a manager (not the assessor, COI-checked) countersigns**. Officers request supplements with a checklist and deadline, across counted rounds; SLA per state shows in the queue, overdue in red. Anti-rubber-stamping threshold: 15 seconds per criterion.

Cross-document checks against cheating

An application can include several documents (form, financial report, CV...). Code extracts and **directly compares like-for-like figures across documents** — total budget, revenue, degree year, page count — and checks for implausible dates. Declaring \$480k while the report says \$1.52m is flagged "mismatch — query", with context quoted from each source.

Outcome letters in the organisation's template

The organisation supplies its standard rejection/feedback letter; the system drafts letters in **exactly that structure, headings and order**, filling in only the signed-off outcomes — replace one file and every letter's format changes.

Layout and table analysis before retrieval

PDF/DOCX layouts are analysed before retrieval: table regions (budgets, staffing...) are kept **row-and-column intact** as one block that can never be split when chunking. Scanned pages without a text layer are detected and flagged "needs OCR" (OCR runs automatically if Tesseract is installed).

Feedback loop → fine-tuning store

At sign-off, every criterion where **the officer overrode the AI** (or decided for it) goes into a ground-truth store with the facts, correct outcome and reason. The store is (1) injected immediately into the judgement prompt as correction examples, and (2) exportable in fine-tuning format for periodic model tuning.

**Conflict-of-interest
check before allocation**

Before an officer takes an application, the system compares officer and application: **same organisation as the applicant → hard block**, the case must be reassigned; same name / named in the application / undeclared affiliation → warning, continuing requires a reason; all of it in the hash-chained log.

Denied-party screening

Every incoming application is fuzzy-matched (misspellings, reordered names, aliases) against banned-persons and sanctions lists, and the organisation is verified against the business register (~20.5 million records, indexed locally). A hit is evidence only — the officer reviews and decides.

CROSS-REFERENCE

Three trust constraints and where they are enforced

CONSTRAINT	MECHANISM	ENFORCED IN	VERIFIED BY
Consistent, but never rubber-stamped	Blind spot-check · per-row confirmation · signing-time check	<code>workflow.py</code> (server)	Spot-check agreement and override rates on the dashboard
Human sign-off is mandatory	DRAFT labels · letters only after full sign-off · separate letter approval	<code>workflow.py</code> (server)	Hash-chained log with one-click integrity check
No penalty for writing style	Judgement pass never sees the source · grammar scoring banned in the prompt	<code>core.py</code> (pipeline)	Bias Lab: 2 application pairs with the same facts in different styles, identical outcomes expected on every criterion; <code>eval.py</code>

GrantLens v2.8 · Criteria quoted verbatim from NSF 22-586 CAREER (nsf.gov) and client-supplied Australian grant guidelines (Wine Australia, Dept. of Industry, NSW RAA, GrantConnect) · Sample applications are synthetic labelled data, not real applications.